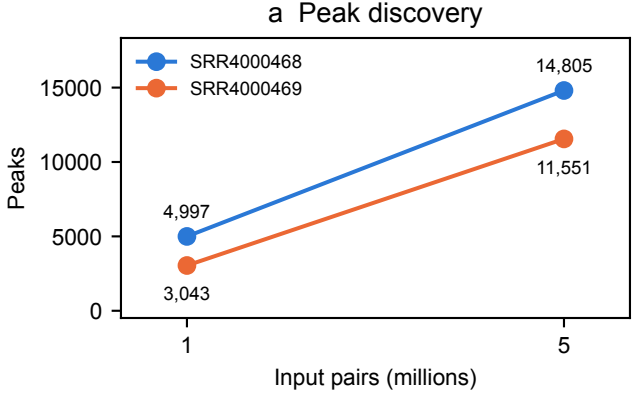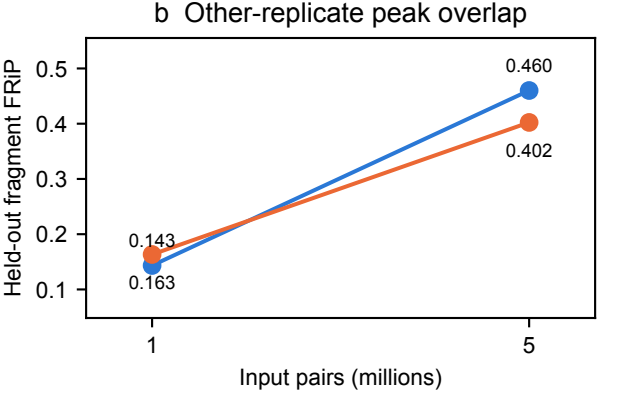Greater depth increases both peak discovery and held-out overlap